ALGEBRA 2 · UNIT 4: POLYNOMIAL FUNCTIONS

Water Slide Designer

Design a smooth water slide with polynomial zeros, multiplicity, and end behavior.

CP2

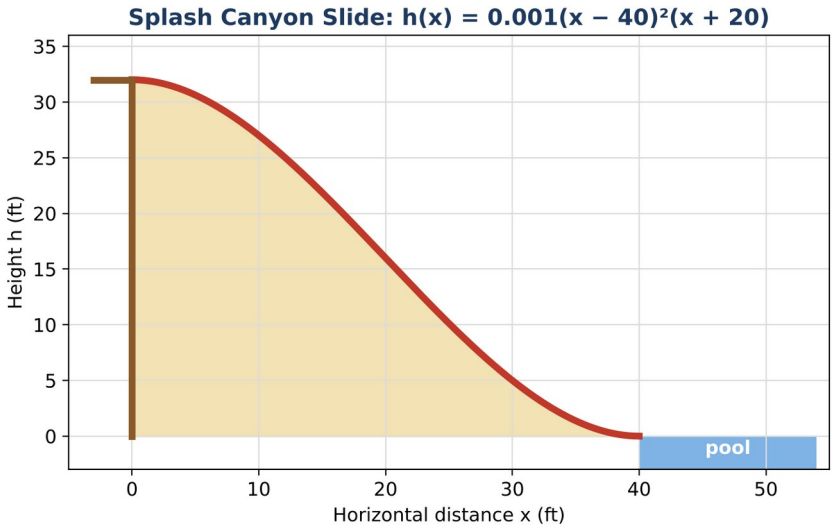
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Splash Canyon Water Park is building a new slide. The profile is $h(x) = 0.001(x - 40)^2(x + 20)$, where x is the horizontal distance in feet and h is the height in feet.

The double zero at x = 40 makes the slide level out smoothly at the pool. You will analyze this design and then create your own.



Learning Goals

- I can evaluate and graph a polynomial function.
- I can relate zeros and multiplicity to the graph.
- I can use synthetic division to check factors.

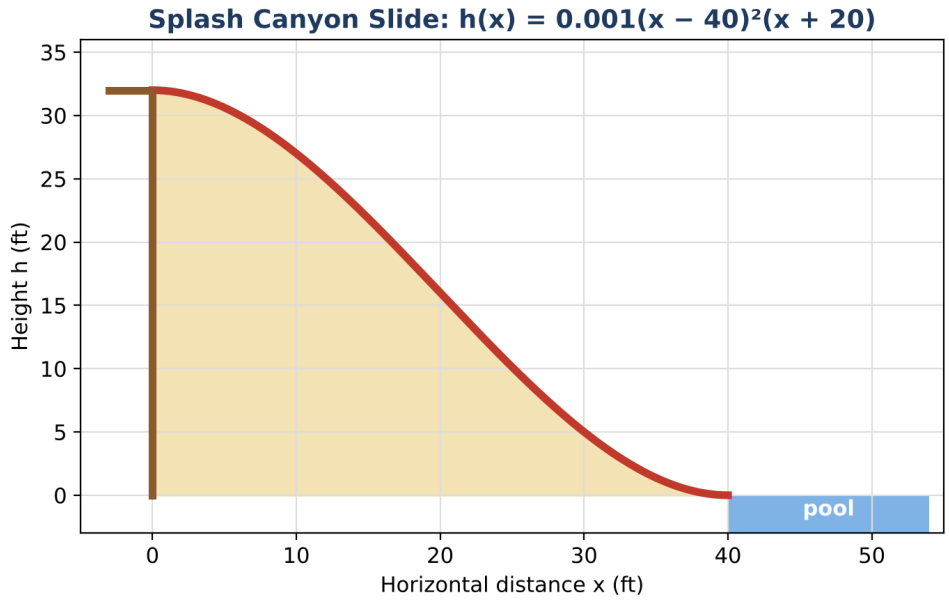
Project Timeline

When	What you will do
Day 1	Read the Slide
Day 2	Check with Division
Day 3	Design a Slide
Final	Slide Design Report + reflection

Part 1: Read the Slide

Day 1 · $h(x) = 0.001(x - 40)^2(x + 20)$

1. Complete the table.



x (ft)	0	10	20	30	40
h(x) (ft)					

2. How tall is the top of the slide? Where does the slide reach the pool?

3. List the zeros of h and the multiplicity of each. Which zero is part of the slide?

4. Explain why a double zero at x = 40 makes a smooth landing.

Part 2: Check with Division

Day 2 · The standard form is $h(x) = 0.001(x^3 - 60x^2 + 32{,}000)$.

5. Use synthetic division to divide $x^3 - 60x^2 + 0x + 32{,}000$ by $(x - 40)$. What is the remainder? What does it tell you?

6. Factor the quotient to get all the factors.

7. Use synthetic division to find the remainder when dividing by $(x - 10)$. Compare $0.001 \times$ remainder with $h(10)$.

Part 3: Design a Slide

Day 3 · Your slide must start 27 ft high and land smoothly 30 ft away.

8. Use the form $h(x) = a(x - 30)^2(x + 15)$. Find a so that $h(0) = 27$.

9. Make a table for $x = 0, 10, 20, 30$.

10. Sketch your slide and compare it with Splash Canyon's.

Final Product: Slide Design Report

Your final product must include:

- Graph of both slides
- Zeros with multiplicity
- Synthetic division work
- Design summary

Reflection

R1. What would a slide with a single zero at the pool look like?

R2. How does synthetic division show that (x − 40) is a factor?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Original slide polynomial starts at the required height and lands smoothly (double zero), with work shown. Goes beyond the minimum with a creative, realistic choice.	Original slide polynomial starts at the required height and lands smoothly (double zero), with work shown.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Report includes the polynomial in factored and standard form, a graph with zeros and multiplicity, division checks, and a design. Polished, organized, and visually engaging.	Report includes the polynomial in factored and standard form, a graph with zeros and multiplicity, division checks, and a design. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.